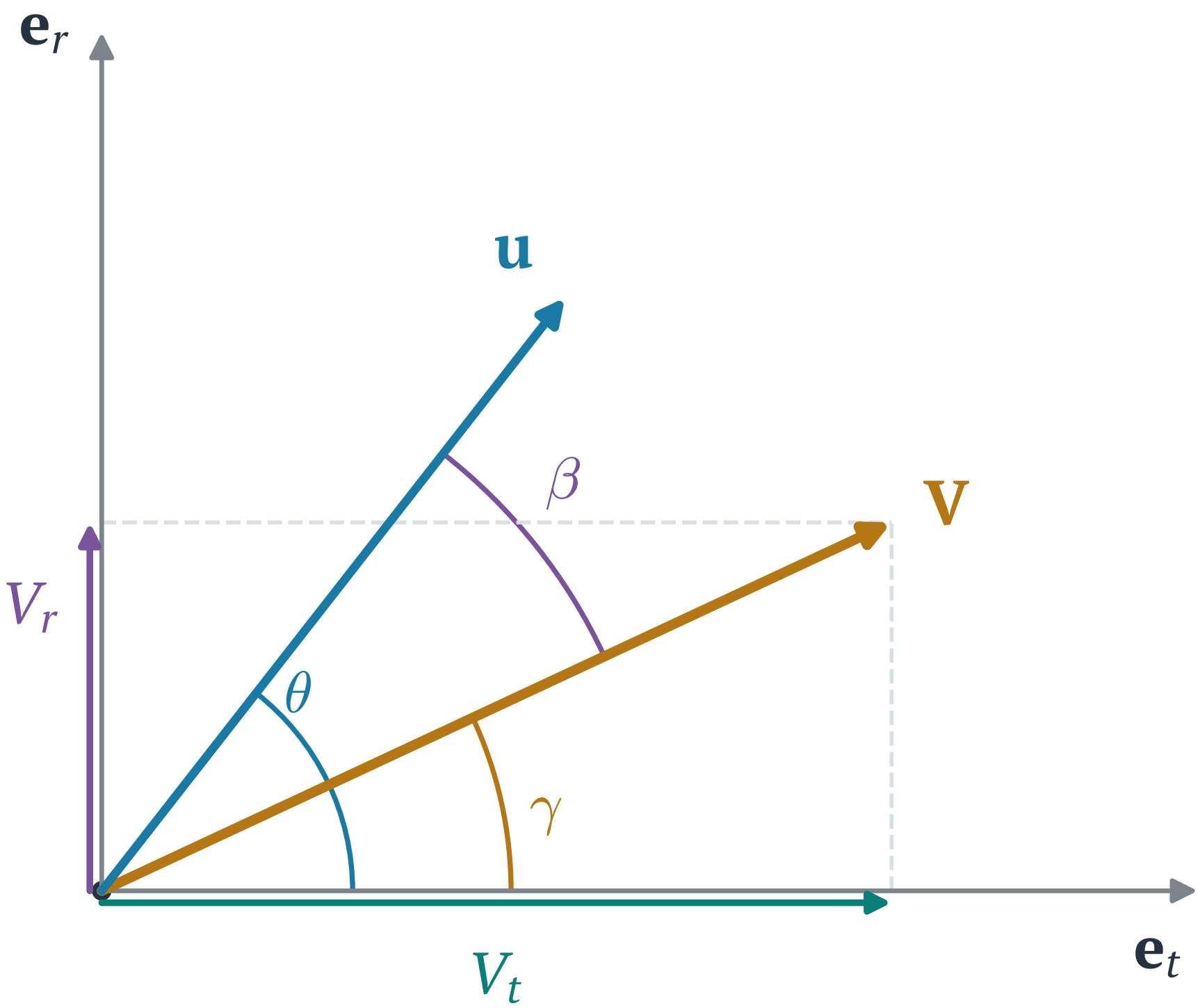

$$\beta = \theta - \gamma, \qquad V_r = V\sin\gamma, \qquad V_t = V\cos\gamma$$